

Student Performance Analysis Dashboard

Project Overview

The Student Performance Analysis Dashboard was developed using Microsoft Excel to analyze factors affecting students' academic performance. The objective of the project was to identify relationships between attendance, study habits, and examination scores while presenting key performance indicators through an interactive dashboard.

Objectives

- Analyze student academic performance using Excel.
- Identify the impact of attendance on examination scores.
- Compare performance across different subjects.
- Evaluate grade distribution among students.
- Present findings through an interactive dashboard for easy decision-making.

Dataset Information

The dataset contains student-related information including:

- Student ID
- Gender
- Attendance Percentage
- Study Hours
- Math Score
- Science Score
- English Score

Additional calculated fields were created:

- Total Score
- Average Score
- Performance Grade
- Attendance Category

Data Cleaning and Preparation

The following data preparation steps were performed:

1. Checked and removed duplicate records.
2. Verified data consistency and formatting.
3. Standardized categorical values.
4. Created calculated columns using Excel formulas.
5. Converted the dataset into a structured table for dynamic analysis.

Tools and Techniques Used

Microsoft Excel Features

- Excel Formulas
- IF Functions
- AVERAGE Function
- PivotTables
- PivotCharts
- Dashboard Design
- Data Aggregation
- KPI Calculations

Dashboard Components

Key Performance Indicators (KPIs)

- Total Students
- Average Score
- Average Attendance Percentage
- Number of Grade A Students

Visualizations

1. Average Score by Gender
2. Average Score by Attendance Category
3. Grade Distribution Analysis
4. Average Subject Score Comparison

Key Findings

1. Attendance and Performance

Students with excellent attendance achieved higher average scores compared to students with poor attendance, indicating a positive relationship between attendance and academic performance.

2. Gender-wise Performance

Female students achieved slightly higher average scores than male students in the analyzed dataset.

3. Grade Distribution

Most students were concentrated in Grade B and Grade C categories, indicating average to above-average academic performance.

4. Subject-wise Analysis

English recorded the highest average score among the three subjects, while Math had the lowest average score.

Recommendations

- Improve attendance monitoring to enhance academic outcomes.
- Provide additional support for students with low attendance.
- Introduce subject-specific improvement programs for weaker subjects.
- Identify at-risk students early using dashboard insights.

Conclusion

The Student Performance Analysis Dashboard successfully transformed raw student data into meaningful insights through Excel-based analysis and visualization. The project demonstrates practical skills in data cleaning, data analysis, dashboard creation, and business reporting using Microsoft Excel.